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# Fair Division

Selected Topics in Algorithms and Optimization (Modul V5C4)

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## Lecture 5: EFX with Charity

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**Outline of this session.** In Lecture 4 we obtained EFX *partial* allocations with high Nash welfare by donating goods; however, many goods may be donated, and agents may envy the donated goods. We then prove the main result of Chaudhury, Kavitha, Mehlhorn and Sgouritsa (SODA 2020): for monotone valuations there is always an EFX allocation in which *fewer than* $n$ goods are donated and *nobody envies* the donated goods. For additive valuations, we can in addition keep the Nash welfare guarantee of Lecture 4. We end with a positive result for complete allocations by Plaut and Roughgarden (2018): if all agents have the *same* valuation, or if there are only two agents, complete EFX allocations exist.

## Partial Allocations and Charity

Throughout, $N = [n]$ is the set of agents and $M$ the set of $m$ goods. Every valuation $v_i : 2^M \to \mathbb{R}_{\geq 0}$ is **monotone** ($S \subseteq T \Rightarrow v_i(S) \leq v_i(T)$) with $v_i(\emptyset) = 0$; additivity is not needed until the section on Nash welfare.

A **partial allocation** is a tuple $X = (X_1, \ldots, X_n)$ of disjoint bundles. The unallocated goods form the **pool** (or *charity*)

$$P \;=\; M \setminus (X_1 \cup \cdots \cup X_n).$$

Envy, EF1 and EFX only compare bundles of agents, so they apply verbatim to partial allocations: $X$ is **EFX** if $v_i(X_i) \geq v_i(X_j \setminus \{g\})$ for all agents $i, j$ and all $g \in X_j$.

**Which partial allocations are reasonable?** The empty allocation ($P = M$) is EFX, so EFX alone says nothing about partial allocations. A reasonable partial allocation should leave only *few* goods unallocated, and these goods should be *unattractive*: if $v_i(P) > v_i(X_i)$, then agent $i$ would rather have the donated goods than her own bundle.

**Theorem 1** (Chaudhury, Kavitha, Mehlhorn, Sgouritsa 2020)**.** *For monotone valuations, there is always a partial allocation $X$ with pool $P$ such that*

*(i) $X$ is EFX;*

*(ii) nobody envies the pool: $v_i(X_i) \geq v_i(P)$ for every agent $i$;*

*(iii) $|P|$ is smaller than the number of sources of the envy graph $G_X$; in particular, $|P| < n$.*

The proof is an algorithm: we start with the empty allocation and apply three update rules as long as possible. The key new concept is a *most envious agent*.

# Tools

## The envy graph (recap)

Recall from Lecture 3 that the **envy graph** $G_X$ has an edge $i \to j$ whenever $v_i(X_i) < v_i(X_j)$, and that a **source** is an agent with no incoming edge (nobody envies her). For a cycle $C$ of $G_X$, let $X^C$ be the allocation obtained by giving every agent on $C$ the bundle of her successor on $C$.

**Lemma 1.** *Let $X$ be EFX and let $C$ be a cycle of $G_X$. Then $X^C$ is EFX and has the same pool, every agent weakly prefers her bundle in $X^C$, agents on $C$ strictly prefer it, and $G_{X^C}$ has fewer edges than $G_X$.*

*Proof.* As in Lecture 3 (where we showed the same for EF1): $X$ and $X^C$ consist of the same bundles, only their owners changed, and the pool is unchanged. Fix an agent $i$ and a bundle $B$ of $X^C$ owned by another agent. If $B$ was $i$'s own bundle in $X$, then $v_i(B) = v_i(X_i) \le v_i(X_i^C)$. Otherwise $B$ was owned by another agent in $X$ too, and EFX of $X$ gives $v_i(B \setminus \{g\}) \le v_i(X_i) \le v_i(X_i^C)$ for all $g \in B$. The statement about the number of edges is part (c) of the rotation lemma in Lecture 3; its proof does not use EF1. □

So, by rotating cycles as long as there are any, we can always make $G_X$ **acyclic**. In an acyclic graph, every agent is reachable from some source (walk backwards along edges until you reach an agent without incoming edges). For an agent $s$, let $R(s)$ denote the set of agents reachable from $s$ in $G_X$, including $s$ itself.

## Minimal envied subsets and most envious agents

**Definition 1** (Envy and strong envy)**.** Let $S \subseteq M$ be a set of goods. Agent $i$ **envies** $S$ if $v_i(S) > v_i(X_i)$, and $S$ is **envied** if some agent envies it. Agent $i$ **strongly envies** $S$ if she still envies $S$ after removing some good from it:

$$v_i(S \setminus \{h\}) \;>\; v_i(X_i) \qquad \text{for some } h \in S.$$

So $X$ is EFX if and only if no agent strongly envies the bundle of another agent. By monotonicity, strong envy implies envy.

**Definition 2** (Minimal envied subset, most envious agent)**.** Let $S \subseteq M$ be envied. A set $Z \subseteq S$ is a **minimal envied subset** of $S$ if $Z$ is envied, but no proper subset of $Z$ is envied by *any* agent:

$$v_j(Z') \le v_j(X_j) \qquad \text{for all agents } j \text{ and all } Z' \subsetneq Z.$$

An agent $i$ with $v_i(Z) > v_i(X_i)$ is called a **most envious agent** of $S$, with *witness* $Z$.

A minimal envied subset always exists: starting from $S$, keep removing goods as long as the remaining set is still envied by somebody (Exercise 2).

**Example 1.** Two agents with additive valuations; the table shows the values of the goods $a, b, c$ of $S = \{a, b, c\}$ and the current values $v_i(X_i)$:

| | $a$ | $b$ | $c$ | $v_i(X_i)$ |
|---|---|---|---|---|
| agent 1 | 2 | 2 | 1 | 3 |
| agent 2 | 1 | 1 | 3 | 2 |

The set $S$ is envied by both agents ($5 > 3$ and $5 > 2$), but it is not a minimal envied subset, since its proper subset $\{c\}$ is envied by agent 2 ($3 > 2$). The minimal envied subsets of $S$ are $\{c\}$ and $\{a, b\}$: agent 1 envies $\{a, b\}$ ($4 > 3$), while no agent envies $\{a\}$ or $\{b\}$. So agent 2 is a most envious agent of $S$ with witness $\{c\}$, and agent 1 is a most envious agent of $S$ with witness $\{a, b\}$. The set $\{a, c\}$ is envied by agent 2 ($4 > 2$), but it is not minimal, since $\{c\}$ is envied. In particular, minimal envied subsets and most envious agents need not be unique.

**Lemma 2** (Key lemma)**.** *Let $Z$ be a minimal envied subset of some envied set. Then for all agents $j$ and all goods $g \in Z$,*

$$v_j(Z \setminus \{g\}) \;\le\; v_j(X_j).$$

*Hence, if afterwards every agent's value is at least as large as before, nobody violates EFX towards the bundle $Z$, whoever owns it.*

*Proof.* $Z \setminus \{g\}$ is a proper subset of $Z$, hence not envied by anyone. □

## The Algorithm

We maintain a partial allocation $X$ with pool $P$ such that

$$\text{(Inv)} \qquad X \text{ is EFX} \quad \text{and} \quad G_X \text{ is acyclic.}$$

Initially $X = (\emptyset, \dots, \emptyset)$ and $P = M$. As long as one of the following rules applies, apply one of them, and afterwards rotate envy cycles until $G_X$ is acyclic again.

**Algorithm 1** (Update rules)**.**

- $\boldsymbol{U_0}$ **(Add a good.)** If there are a good $g \in P$ and an agent $i$ such that giving $g$ to $i$ keeps $X$ EFX, do so: $X_i \leftarrow X_i \cup \{g\}$ and $P \leftarrow P \setminus \{g\}$.
- $\boldsymbol{U_1}$ **(Take from the pool.)** If some agent envies $P$, let $Z$ be a minimal envied subset of $P$ and $i$ a most envious agent of $P$ with witness $Z$. Agent $i$ swaps her bundle for $Z$:

$$X_i \leftarrow Z, \qquad P \leftarrow (P \setminus Z) \cup X_i^{\text{old}}.$$

- $\boldsymbol{U_2}$ **(Improving configuration.)** If there is an improving configuration (defined below), execute it.

Rule $U_0$ is the greedy step. Rule $U_1$ makes sure that in the end nobody envies the pool. Rule $U_2$ makes progress when the pool is large but no good can be added directly; Lemma 6 below shows how to find an improving configuration in that case.

### Rule $U_0$

**Lemma 3.** *Rule $U_0$ keeps $X$ EFX, does not decrease any agent's value, and decreases $|P|$ by one.*

*Proof.* EFX holds by the definition of the rule, and by monotonicity no value decreases. □

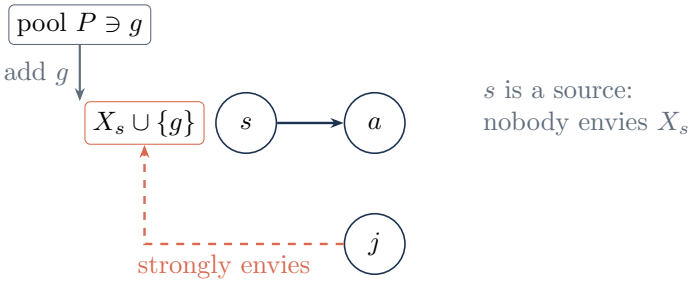
### Rule $U_1$

**Lemma 4.** *Rule $U_1$ keeps $X$ EFX, strictly increases the value of agent $i$, and does not change the other bundles.*

*Proof.* Only $X_i$ changes, and $v_i(Z) > v_i(X_i)$ since $i$ is a most envious agent with witness $Z$. Agent $i$ towards another bundle $X_k$: $v_i(X_k \setminus \{g\}) \le v_i(X_i) < v_i(Z)$. Another agent $j$ towards the new bundle $Z$: $v_j(Z \setminus \{g\}) \le v_j(X_j)$ by Lemma 2. All other comparisons are unchanged. □

### What do we learn when $U_0$ does not apply?

Let $s$ be a source of $G_X$ and $g \in P$. If we give $g$ to $s$, then $s$ herself is happy, and nothing changes between the other agents. So if $U_0$ does not apply, the only obstacle is that some other agent $j$ strongly envies the new bundle $X_s \cup \{g\}$, i.e. violates EFX towards it.

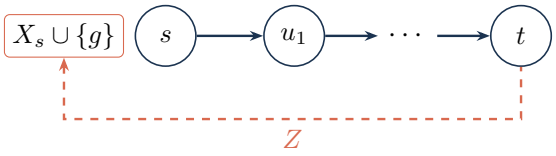



**Observation 1.** *Suppose that $U_0$ does not apply, and let $s$ be a source of $G_X$. Then for every $g \in P$, the set $X_s \cup \{g\}$ is envied by some agent $j \neq s$.*

*Proof.* Giving $g$ to $s$ creates no EFX violation of $s$ herself (her value does not decrease) and does not affect comparisons between other agents. Since $U_0$ does not apply, some agent $j \neq s$ strongly envies $X_s \cup \{g\}$: there is $h \in X_s \cup \{g\}$ with $v_j\big((X_s \cup \{g\}) \setminus \{h\}\big) > v_j(X_j)$. By monotonicity, $v_j(X_s \cup \{g\}) > v_j(X_j)$, so $j$ envies $X_s \cup \{g\}$. □

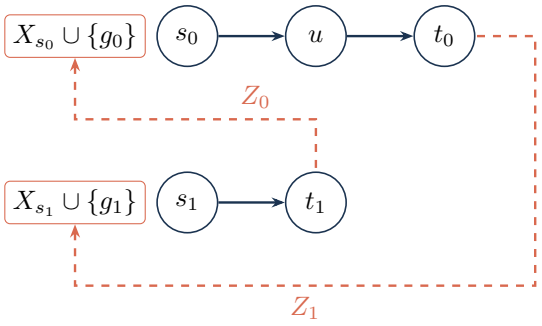
So $X_s \cup \{g\}$ is envied, and it has a most envious agent $t$ with a witness $Z \subseteq X_s \cup \{g\}$ (the agent $t$ need not be $j$). Agent $t$ would like to receive $Z$. There are two cases.

**Case 1: $t$ is reachable from $s$.** Take a path $s = u_0 \to u_1 \to \cdots \to u_q = t$ in $G_X$. Together with the dashed edge "$t$ envies $Z$", it closes a cycle, and we can rotate along it as in envy-cycle elimination:




Every $u_p$ with $p < q$ takes the bundle $X_{u_{p+1}}$ of her successor, $t$ takes $Z$, and the goods of $(X_s \cup \{g\}) \setminus Z$ go to the pool. Every agent on the cycle strictly gains: $u_p$ envies $u_{p+1}$, and $t$ envies $Z$. As in Lemma 1, nobody violates EFX towards a bundle that only moved, and by the key lemma (Lemma 2) nobody violates EFX towards $Z$. (If $t = s$, then $q = 0$ and $s$ simply replaces $X_s$ by $Z$.)

**Case 2: $t$ is not reachable from $s$.** Since $G_X$ is acyclic, $t$ is reachable from *another* source $s'$. Now take a second good $g' \in P$: the set $X_{s'} \cup \{g'\}$ is envied by Observation 1, so it has a most envious agent $t'$, with witness $Z'$. If $t'$ is reachable from $s$, a cycle closes again, now through both sources. (If $t'$ is reachable from $s'$, we are in Case 1 for $s'$ alone.) In the picture below, $s_0 = s$, $s_1 = s'$, $t_1 = t$ and $t_0 = t'$: the most envious agent of $X_{s_0} \cup \{g_0\}$ is $t_1 \in R(s_1)$, and the most envious agent of $X_{s_1} \cup \{g_1\}$ is $t_0 \in R(s_0)$.




Agent $s_0$ takes $X_u$, $u$ takes $X_{t_0}$, and $t_0$ takes $Z_1$; agent $s_1$ takes $X_{t_1}$, and $t_1$ takes $Z_0$. Again every agent on the cycle strictly gains. If $t'$ is reachable from neither $s$ nor $s'$, then it is reachable from a third source $s''$, and we continue with $X_{s''} \cup \{g''\}$ for a third good $g''$, and so on. Each round uses a *new* source, so after at most (number of sources) rounds a cycle closes, provided that the pool contains enough goods: one for every source we use.

## Improving configurations

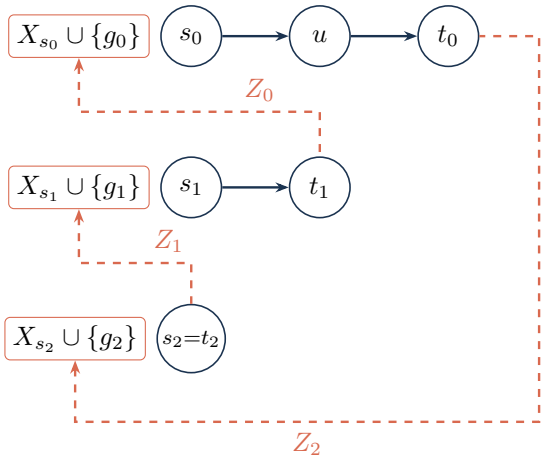
We now make this precise. A cycle as above is called an improving configuration; Rule $U_2$ executes it.

**Definition 3** (Improving configuration)**.** An **improving configuration** consists of $\ell \geq 1$ distinct sources $s_0, \ldots, s_{\ell-1}$ of $G_X$, distinct goods $g_0, \ldots, g_{\ell-1} \in P$, and agents $t_0, \ldots, t_{\ell-1}$ such that, with indices modulo $\ell$:

- $t_{r+1}$ is a most envious agent of $X_{s_r} \cup \{g_r\}$, with witness $Z_r \subseteq X_{s_r} \cup \{g_r\}$;
- there are vertex-disjoint paths $\pi_0, \ldots, \pi_{\ell-1}$ in $G_X$, where $\pi_r$ goes from $s_r$ to $t_r$ (a path may have length 0, i.e. $t_r = s_r$).

**Executing** it means: along every path $\pi_r = (s_r = u_0 \to u_1 \to \cdots \to u_q = t_r)$, agent $u_p$ receives the bundle $X_{u_{p+1}}$ for $p < q$, and agent $t_r$ receives $Z_{r-1}$. All goods of $\big(X_{s_r} \cup \{g_r\}\big) \setminus Z_r$ go to the pool.

Case 1 above is an improving configuration with $\ell = 1$, and the last picture shows one with $\ell = 2$. In general, the sources, the sets $X_{s_r} \cup \{g_r\}$ and the paths look as follows (here $\ell = 3$):




Executing it: $s_0$ takes $X_u$, $u$ takes $X_{t_0}$ and $t_0$ takes $Z_2$; $s_1$ takes $X_{t_1}$ and $t_1$ takes $Z_0$; and $s_2 = t_2$, whose path has length 0, takes $Z_1$. The remaining goods of the sets $X_{s_r} \cup \{g_r\}$ go to the pool.

**Lemma 5.** *Executing an improving configuration keeps $X$ EFX, weakly increases the value of every agent, and strictly increases the value of every agent on the paths $\pi_0, \dots, \pi_{\ell-1}$.*

*Proof. Well defined.* The paths are vertex-disjoint and each of them ends in exactly one $t_r$, so every agent receives at most one new bundle. The bundles handed on along $\pi_r$ belong to agents $u_1, \dots, u_q$ of $\pi_r$ other than $s_r$; none of them is a source of the configuration (each $s_{r'}$ is the first vertex of its own path), so these bundles are not used in any $Z_{r'}$. The sets $Z_0, \dots, Z_{\ell-1}$ are disjoint, since they lie in the disjoint sets $X_{s_r} \cup \{g_r\}$. So the new bundles are disjoint.

*Values.* An agent $u_p$ with $p < q$ envies $u_{p+1}$, so she strictly gains. Agent $t_r$ receives $Z_{r-1}$ and is a most envious agent with witness $Z_{r-1}$, so $v_{t_r}(Z_{r-1}) > v_{t_r}(X_{t_r})$. Agents not on any path keep their bundles.

*EFX.* Let $X'$ be the new allocation. Its bundles are of two kinds: old bundles $X_u$ (of agents $u$ that are not among the sources $s_r$), and the sets $Z_r$. Let $j$ be an agent and $B$ a bundle of $X'$ held by another agent. If $B = X_j$ was $j$'s own old bundle, then $v_j(B) = v_j(X_j) \le v_j(X'_j)$. If $B$ is another old bundle, EFX of $X$ gives $v_j(B \setminus \{g\}) \le v_j(X_j) \le v_j(X'_j)$. If $B = Z_r$, Lemma 2 gives $v_j(Z_r \setminus \{g\}) \le v_j(X_j) \le v_j(X'_j)$. □

### Large pools admit an improving configuration

**Lemma 6.** *Suppose that $U_0$ does not apply, $G_X$ is acyclic, and $|P|$ is at least the number of sources of $G_X$. Then there is an improving configuration.*

*Proof.* We formalise the procedure from Cases 1 and 2. Start with any source $s_0$ and any good $g_0 \in P$. For $r = 1, 2, \dots$:

- By Observation 1, $X_{s_{r-1}} \cup \{g_{r-1}\}$ is envied. Let $t_r$ be a most envious agent of it, with witness $Z_{r-1}$.
- If $t_r \in R(s_0) \cup \dots \cup R(s_{r-1})$, stop. Otherwise, let $s_r$ be a source with $t_r \in R(s_r)$ (it exists since $G_X$ is acyclic), and let $g_r \in P$ be a good different from $g_0, \dots, g_{r-1}$.

The new source $s_r$ is different from $s_0, \dots, s_{r-1}$, because $t_r \in R(s_r)$ but $t_r \notin R(s_{r'})$ for $r' < r$. Hence the process stops after at most (number of sources) steps, and we never run out of goods, since $|P|$ is at least the number of sources.

Suppose we stop in step $k$, and let $j < k$ be an index with $t_k \in R(s_j)$. We use the sources $s_j, \dots, s_{k-1}$ with the goods $g_j, \dots, g_{k-1}$ and the following paths:

- for $j < r < k$, a path $\pi_r$ from $s_r$ to $t_r$ (it exists since $t_r \in R(s_r)$);
- a path $\pi_j$ from $s_j$ to $t_k$; it closes the cycle, since $t_k$ is a most envious agent of $X_{s_{k-1}} \cup \{g_{k-1}\}$.

After re-indexing from 0, this has the form of an improving configuration. It remains to show that the paths are vertex-disjoint. For $j < r < k$, the path $\pi_r$ avoids $R(s_{r'})$ for every $r' < r$: if a vertex $u$ of $\pi_r$ were reachable from $s_{r'}$, then so would be $t_r$ (via the rest of $\pi_r$), contradicting the fact that we did not stop in step $r$. Since $\pi_{r'} \subseteq R(s_{r'})$ for every $j \le r' < r$, the path $\pi_r$ is disjoint from all paths with a smaller index. □

## Proof of the Main Theorem

*Proof of Theorem 1. Invariant.* The empty allocation is EFX and $G_X$ has no edges. By Lemmas 1, 3, 4 and 5, every rule and every rotation keeps $X$ EFX, and after the rotations $G_X$ is acyclic again (each rotation removes edges).

*Termination.* Let $\varphi(X) = \sum_i v_i(X_i)$ be the social welfare. Every rotation and every application of $U_1$ or $U_2$ strictly increases $\varphi(X)$. Rule $U_0$ does not decrease $\varphi(X)$ and decreases $|P|$. Hence the pair $\big(\varphi(X), -|P|\big)$ strictly increases in lexicographic order with every step. There are

only finitely many partial allocations, so no partial allocation is visited twice, and the algorithm terminates.

*Output.* When no rule applies:

(i) $X$ is EFX by the invariant;

(ii) $U_1$ does not apply, so nobody envies $P$;

(iii) $U_0$ and $U_2$ do not apply, so by Lemma 6, $|P|$ is smaller than the number of sources of $G_X$, which is at most $n$. □

## Charity and Nash Welfare

The proof shows more than Theorem 1: no step of the algorithm makes any agent worse off.

**Observation 2.** *Every rotation and every application of $U_0$, $U_1$ and $U_2$ weakly increases the value of every agent.*

*Proof.* Lemma 1 for rotations, Lemmas 3 and 4 for $U_0$ and $U_1$ (in $U_1$ only agent $i$'s bundle changes), and Lemma 5 for $U_2$. □

Nothing in the proof used that we start from the empty allocation; we only need the invariant (Inv). So we may start from *any* EFX partial allocation $X^0$ (after rotating envy cycles, which keeps EFX and makes nobody worse off), and the output $X$ then satisfies $v_i(X_i) \geq v_i(X_i^0)$ for every agent $i$. Combining this with Lecture 4 gives:

**Theorem 2** (Chaudhury, Kavitha, Mehlhorn, Sgouritsa 2020)**.** *For additive valuations, let $X^*$ be an MNW allocation. There is a partial allocation $X$ with pool $P$ that satisfies (i)–(iii) of Theorem 1 and $v_i(X_i) \geq \frac{1}{2} v_i(X_i^*)$ for every agent $i$. In particular,* $\mathrm{NW}(X) \geq 2^{-(1-1/n)} \operatorname{opt} > \frac{1}{2} \operatorname{opt}$.

*Proof.* By Lecture 4 (Caragiannis, Gravin, Huang 2019), there is an EFX partial allocation $Y$ with $Y_i \subseteq X_i^*$, $v_i(Y_i) \geq \frac{1}{2} v_i(X_i^*)$ for all $i$, and $Y_{i_0} = X_{i_0}^*$ for some agent $i_0$. Start the algorithm from $Y$. By Observation 2, the output $X$ satisfies $v_i(X_i) \geq v_i(Y_i)$ for all $i$, and it satisfies (i)–(iii) as in the proof of Theorem 1. Hence

$$\mathrm{NW}(X)^n \;\geq\; \prod_i v_i(Y_i) \;\geq\; 2^{-(n-1)} \prod_i v_i(X_i^*) \;=\; 2^{-(n-1)} \operatorname{opt}^n. \qquad \square$$

So, compared with Lecture 4, we keep EFX and the Nash welfare guarantee, but now fewer than $n$ goods are donated and nobody envies them.

## EFX for Identical Valuations

We return to *complete* allocations. In this section, all agents have the *same* monotone valuation $v$.

The following simple local-search algorithm finds an EFX allocation. For a bundle $B$, a **smallest good** of $B$ is a good $g \in B$ that maximises $v(B \setminus \{g\})$ (for additive $v$: a good of minimum value). An agent $i$ is a **minimum agent** if $v(X_i)$ is minimum.

**Algorithm 2** (EFX for identical valuations)**.** Start with an arbitrary complete allocation $X$. While $X$ is not EFX:

- take a bundle $X_j$ that some agent strongly envies, i.e. $v(X_h) < v(X_j \setminus \{g'\})$ for some agent $h$ and some $g' \in X_j$, and let $g$ be a smallest good of $X_j$;

- take a minimum agent $i$;
- move $g$ from $X_j$ to $X_i$.

Clearly, if the algorithm stops, then $X$ is EFX. Why does it stop?

**Theorem 3** (Plaut, Roughgarden 2018)**.** *For identical monotone valuations, Algorithm 2 terminates. Hence complete EFX allocations exist.*

*Proof.* Consider a move, and let $X'$ be the new allocation: $X'_i = X_i \cup \{g\}$, $X'_j = X_j \setminus \{g\}$, and $X'_\ell = X_\ell$ for all other agents $\ell$. Recall that $v(X_h) < v(X_j \setminus \{g'\})$ for some agent $h$ and some $g' \in X_j$.

*The minimum value never decreases.* Agent $i$ does not lose value, by monotonicity. For agent $j$,

$$v(X'_j) \;=\; v(X_j \setminus \{g\}) \;\geq\; v(X_j \setminus \{g'\}) \;>\; v(X_h) \;\geq\; v(X_i),$$

by the choice of $g$ as a smallest good and of $i$ as a minimum agent. In particular $v(X_j) > v(X_i)$, so $j \neq i$. All other bundles are unchanged. Hence the minimum value of $X'$ is at least that of $X$.

*What if the minimum does not change?* This can happen, for instance if several agents have the minimum value, or if $v(X_i \cup \{g\}) = v(X_i)$ (for additive $v$: $v(g) = 0$). So we refine the minimum value as follows. For a bundle $B$, let $\kappa(B) = (v(B), |B|)$; such pairs are compared lexicographically. For an allocation $X$, let $\kappa(X) = (\kappa_1, \ldots, \kappa_n)$ be the pairs $\kappa(X_1), \ldots, \kappa(X_n)$ sorted in non-decreasing order. We show that $\kappa(X')$ is lexicographically larger than $\kappa(X)$.

- $\kappa(X'_i) > \kappa(X_i)$: the value does not decrease and the size grows.
- $\kappa(X'_j) > \kappa(X_i)$ and $\kappa(X_j) > \kappa(X_i)$: both values are at least $v(X_j \setminus \{g\}) > v(X_i)$.

Hence $X$ and $X'$ have exactly the same bundles with $\kappa < \kappa(X_i)$, and $X'$ has one bundle with $\kappa = \kappa(X_i)$ fewer than $X$ (the bundle $X_i$ is gone, and no new bundle with $\kappa \leq \kappa(X_i)$ appears). Sort both sequences: they agree on all entries smaller than $\kappa(X_i)$ and on all but one of the entries equal to $\kappa(X_i)$; at the next position, $\kappa(X)$ has the entry $\kappa(X_i)$, while $\kappa(X')$ has an entry larger than $\kappa(X_i)$. So $\kappa(X')$ is lexicographically larger than $\kappa(X)$.

Since there are finitely many allocations, no allocation is visited twice, and the algorithm terminates. When it stops, $X$ is EFX. □

*Leximin++.* An allocation maximising $\kappa(X)$ is called **leximin++**. By the proof, no move is possible from it, so every leximin++ allocation is EFX.

**Corollary 1** (Plaut, Roughgarden 2018)**.** *For two agents with (possibly different) monotone valuations, a complete EFX allocation exists.*

*Proof.* Cut and choose. By Theorem 3, there is an EFX allocation $(B_1, B_2)$ for two agents who both have valuation $v_1$: $v_1(B_1) \geq v_1(B_2 \setminus \{g\})$ for all $g \in B_2$ and $v_1(B_2) \geq v_1(B_1 \setminus \{g\})$ for all $g \in B_1$. Agent 2 takes her favourite of $B_1, B_2$, and agent 1 gets the other one. Agent 2 envies nobody, and agent 1 satisfies EFX whichever bundle she gets. □

**Remarks.**

- *Computation.* All steps only use value queries $v_i(S)$; a minimal envied subset of a set $S$ can be found with $O(n|S|)$ queries (Exercise 2). The number of iterations is bounded via the social welfare, so the algorithm runs in pseudo-polynomial time: for integer valuations, polynomial in $n$, $m$ and $V = \max_i v_i(M)$ (Exercise 3).
- *Identical valuations.* If all agents have the same valuation, the algorithm ends with $P = \emptyset$ (Exercise 1). This gives another proof of Theorem 3.

- *Less charity.* For additive valuations, later work reduced the number of donated goods: to at most $n-2$, again with nobody envying them (Berger, Cohen, Feldman, Fiat 2022), and to a sublinear number of goods if one relaxes EFX to $(1-\varepsilon)$-EFX (Chaudhury, Garg, Mehlhorn, Mehta, Misra 2021). For general monotone valuations, some charity is unavoidable, since complete EFX allocations need not exist (Akrami, Mayorov, Mehlhorn, Srinivas, Weidenbach 2026; Mackenzie, Suzuki 2026). For additive valuations, whether $P = \emptyset$ is always possible is the open EFX problem.

## Exercises

**Question 1 (Identical valuations).** Suppose all agents have the same monotone valuation $v$. Show that whenever $P \neq \emptyset$, one of the rules $U_0$, $U_1$, $U_2$ applies. Conclude that a complete EFX allocation exists (another proof of Theorem 3).

**Question 2 (Computing a minimal envied subset).** Let $S$ be an envied set. Go through the goods of $S$ in any order, and remove the current good whenever the remaining set is still envied by some agent. Show that the result is a minimal envied subset of $S$, and that $O(n|S|)$ value queries suffice.

**Question 3 (Running time).** Consider the variant of the algorithm in which $U_2$ is applied only if $U_0$ does not apply and $|P|$ is at least the number of sources of $G_X$, using the improving configuration constructed in the proof of Lemma 6.

(a) Show that this variant still proves Theorem 1.

(b) Suppose all values $v_i(S)$ are integers, and let $V = \max_i v_i(M)$. Show that the variant performs at most $(nV+1)(m+1)$ applications of $U_0$, $U_1$ and $U_2$, and that each application, including the subsequent rotations, needs a number of value queries polynomial in $n$ and $m$.